

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The functions f and g are defined by the given equations, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the maximum value of the function it defines, where $x \geq 0$?

- I. $f(x) = 33(0.4)^{x+3}$
- II. $g(x) = 33(0.16)(0.4)^{x-2}$

Answer

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

Correct Answer: B

Rationale

Choice B is correct. Functions f and g are both exponential functions with a base of 0.40 . Since 0.40 is less than 1 , functions f and g are both decreasing exponential functions. This means that $f(x)$ and $g(x)$ decrease as x increases. Since $f(x)$ and $g(x)$ decrease as x increases, the maximum value of each function occurs at the least value of x for which the function is defined. It's given that functions f and g are defined for $x \geq 0$. Therefore, the maximum value of each function occurs at $x = 0$. Substituting 0 for x in the equation defining f yields $f(0) = 33(0.4)^{0+3}$, which is equivalent to $f(0) = 33(0.4)^3$, or $f(0) = 2.112$. Therefore, the maximum value of f is 2.112 . Since the equation $f(x) = 33(0.4)^{x+3}$ doesn't display the value 2.112 , the equation defining f doesn't display the maximum value of f . Substituting 0 for x in the equation defining g yields $g(0) = 33(0.16)(0.4)^{0-2}$, which can be rewritten as $g(0) = 33(0.16)\left(\frac{1}{0.4^2}\right)$, or $g(0) = 33(0.16)\left(\frac{1}{0.16}\right)$, which is equivalent to $g(0) = 33$. Therefore, the maximum value of g is 33 . Since the equation $g(x) = 33(0.16)(0.4)^{x-2}$ displays the value 33 , the equation defining g displays the maximum value of g . Thus, only equation II displays, as a constant or coefficient, the maximum value of the function it defines.

- Choice A is incorrect and may result from conceptual or calculation errors.
- Choice C is incorrect and may result from conceptual or calculation errors.
- Choice D is incorrect and may result from conceptual or calculation errors.